

Sagnik Chatterjee

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Education and Employment

- DEC 2025 – Present 📖 **Visiting Fellow (Postdoctoral Researcher)** at STCS, TIFR Mumbai.
Host: Prof. Jatin Batra.
- Jul 2019 – Jun 2025 📖 **Ph.D.** in Computer Science and Engineering at IIIT-Delhi.
Advisor: Prof. Debajyoti Bera.
- SEP 2017 – MAR 2019 📖 **Staff Consultant**, Oracle Financial Services Software Limited, Bangalore.
- Aug 2013 – Aug 2017 📖 **B.Tech.** in Computer Science and Engineering at MAKAUT (formerly, WBUT-Inhouse).
GPA: 8.10/10.

Research

Ph.D. Thesis

- 1 **S. Chatterjee**, “Designing quantum learning algorithms for classical objects,” Available at <https://repository.iiitd.edu.in/xmlui/handle/123456789/1758>, PhD thesis, IIIT-Delhi, May 2025.

Journal Articles

- 1 Y. Saxena, T. Sapv, **S. Chatterjee**, and R. K. Lee, “Generalized cross-resonance scheme for maximally-entangling two-qutrit gates,” *Physical Review Research*, Jun. 2026. 📄 DOI: 10.1103/qqtg-x6jg.
- 2 **S. Chatterjee**, R. Bhatia, P. S. Chani, and D. Bera, “Quantum boosting using domain-partitioning hypotheses,” *Quantum Machine Intelligence*, vol. 5, no. 2, pp. 1–20, 2023. 📄 DOI: 10.1007/S42484-023-00122-3.

Conference Proceedings

- 1 S. Das[†], **S. Chatterjee**[†], and J. Batra, “Convex basins in single-index model loss landscapes: Applications to robust recovery under strong adversarial corruption,” in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. 📄 URL: <https://arxiv.org/abs/2605.29497>.
- 2 M. Mukherjee, **S. Chatterjee**, and A. Sethi, “Perfect secret key generation for a class of hypergraphical sources,” in *IEEE International Symposium on Information Theory (ISIT)*, 2026. 📄 URL: <https://arxiv.org/abs/2601.10697>.
- 3 **S. Chatterjee**[†], M. Mukherjee[†], and A. Sethi[†], “Generalization bounds for dependent data using online-to-batch conversion,” in *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. 📄 URL: <https://proceedings.mlr.press/v258/chatterjee25b.html>.
- 4 **S. Chatterjee**, T. SAPV, and D. Bera, “Efficient quantum agnostic improper learning of decision trees,” in *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024. 📄 URL: <https://proceedings.mlr.press/v238/chatterjee24a.html>.

[†] denotes equal contribution.

Preprints, Workshop Papers, and Posters

- 1 **S. Chatterjee**, *The Quantum Learning Menagerie (A survey on Quantum learning for Classical concepts)*, 2026. 📄 URL: <https://arxiv.org/abs/2602.01054>.
- 2 **S. Chatterjee** and V. Kungurtsev, *Quantum solutions to the privacy vs. utility tradeoff*, 2023. arXiv: 2307.03118.
- 3 **S. Chatterjee** and D. Bera, *Applying the quantum alternating operator ansatz to the graph matching problem*, Extended Abstract at the The 20th Asian Quantum Information Science (AQIS) Conference, 2020. arXiv: 2011.11918.

Teaching

Instructor (Short Courses)

- Mar 2026  Designed and taught a mini-course (6 lectures) at **IIT (ISM) Dhanbad**.
Topic: axioms of quantum computing, quantum query speedups for promise problems, variational quantum algorithms, and hands-on introduction to QISKIT.
- Feb 2024  **Designed and taught 3 lectures** at the **Chennai Mathematical Institute**.
Topic 1: The Harrow-Hassidim-Lloyd Algorithm and its extensions.
Topic 2: Block Encoding and Linear Combination of Unitaries.
- Aug 2022  Designed and taught 3 lectures for a refresher module at **IIIT-Delhi**.
Topic: C programming for Operating Systems.

Teaching Assistantship (IIITD)

- Theory of Computation  Winter 2020, Winter 2021, Winter 2024. **[Head TA]**
- Data Structures  Summer 2022. **[Head TA]**
- Intro to Quantum Computing  Winter 2023. **[Sole TA]**
- Modern Algorithm Design  Monsoon 2020, Monsoon 2021. **[Sole TA]**

My duties included designing and conducting tutorials, creating assignments, and holding office hours and remedial sessions for all the above courses.

Research Visits and Internships

- AUG 2024– SEP 2024  **ACMU, Indian Statistical Institute**, Kolkata.
Host: Prof. Sourav Chakraborty.
- JUL 2023– AUG 2023  **Czech Technical University**, Prague.
Host: Prof. Jakub Marecek, Prof. Vyacheslav Kungurtsev.
- FEB 2017 – APR 2017  **Systems Engineering Intern**, Infosys Limited.
- JUN 2016– JUL 2016  **Data-Science Intern**, AICircle Pte Ltd.

Invited Talks

- Oct 2025  STCS seminar, TIFR Mumbai.
Generalization bounds for dependent data using online-to-batch conversion.
- May 2025  CS Colloquium, Ashoka University.
Efficient quantum agnostic improper learning of decision trees.
- Mar 2025  Quantum Computing Workshop by Edunautic, IIT Delhi.
Full day workshop from fundamentals to hands-on instruction.
- Jan 2025  Young Scientists Session at QAC 2025 symposium, C-DAC and DIAT Pune.
Modern Algorithmic Primitives in Quantum Computing.
- Oct 2024  ACMU seminar, ISI Kolkata.
Generalization bounds for dependent data using online-to-batch conversion.
- July 2024  Recent Trends in Algorithms Workshop 2024.
Efficient quantum agnostic improper learning of decision trees.
- June 2024  ACMU seminar, ISI Kolkata.
Efficient quantum agnostic improper learning of decision trees.
- February 2024  Quantum Computing Semester, Chennai Mathematical Institute.
Quantum Algorithms for Linear Algebra.
Block Encodings and Linear Combination of Unitaries.
- September 2023  Center for Quantum Computing Science, University of Latvia.
Efficient quantum agnostic improper learning of decision trees.

Invited Talks (continued)

	IDA Seminar, Czech Technical University. <i>Efficient quantum agnostic improper learning of decision trees.</i>
November 2022	Short Talk at the 6th Quantum Techniques in Machine Learning (QTML 2022). <i>Quantum boosting using domain-partitioning hypotheses.</i>
March 2022	Theory Seminar, Indian Institute of Information Technology Delhi. <i>Quantum boosting using domain-partitioning hypotheses.</i>
December 2020	Faculty Development Programme, JNTU Anantapur. <i>Quantum Machine Learning.</i>

Miscellaneous

Awards and Achievements

MAY 2026	Recognized as a Gold Reviewer at ICML 2026 .
FEB 2026	Invited poster at ACM ARCS 2026 .
FEB 2025	Invited poster at ACM ARCS 2025 . (Declined)
AUG 2024	IIITD Dean's List for Best Teaching Assistant (Theory of Computation).
APR 2024	AISTATS 2024 Registration Grant.
JAN 2022	QIP 2022 Student Travel Award. (Not availed due to COVID restrictions)
FEB 2020	Runners Up, IBMQ Awards in the <i>Teach Me Quantum 2019</i> category. (4000 USD cash prize)
OCT 2019	Full Scholarship to attend the 4th Winter School in CSE organized by the IIAS, HUJI, Jerusalem.

Service

Conference Reviewing	AAAI ('26); FSTTCS ('26); ICML ('26,'25); ICLR ('26,'25); AISTATS ('26,'25); STACS ('25); NeurIPS ('25, '24).
Journal Reviewing	Scientific Reports ('24); Quantum ('24, '25).
Organisation	Co-organiser and Webadmin for the QISE workshop at FSTTCS 2021; Organised a semester long colloquium: Talks on Quantum Computing.

References

Available on Request